

ALEKSANDROVSKOE, Russian Federation

WMO: 239550

Lat: 60.433N

Lon: 77.867E

Elev: 159

StdP: 14.61

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.3	-36.0	-45.6	0.4	-40.5	-40.4	0.5	-35.6	16.4	4.2	14.9	5.2	4.1	290	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	82.3	66.9	79.3	65.4	76.1	64.0	69.2	78.3	67.3	76.0	65.4	73.8	6.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	96.0	73.7	63.9	89.4	72.0	61.9	83.4	70.3	33.5	78.7	31.9	75.9	30.5	73.8	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.5	13.2	DB	-44.0	87.1	7.4	2.3	-49.3	88.8	-53.6	90.2	-57.7	91.5	-63.1	93.1	(4)
(5)			WB	-43.6	73.4	6.5	3.3	-48.2	75.7	-52.1	77.6	-55.7	79.4	-60.4	81.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.9	-6.4	0.0	17.2	30.3	44.8	59.8	64.0	57.7	46.3	32.3	10.7	-0.3		
	DBStd	27.12	17.06	15.60	12.88	11.73	10.66	9.20	6.71	6.83	8.02	9.70	16.26	17.15		
	HDD50	8483	1749	1400	1017	592	232	23	1	11	168	550	1180	1560		
	HDD65	12965	2214	1820	1482	1040	630	205	101	243	563	1012	1630	2025		
	CDD50	1136	0	0	0	2	72	317	436	251	55	3	0	0		
	CDD65	144	0	0	0	0	5	49	71	18	1	0	0	0		
	CDH74	1111	0	0	0	0	63	390	520	131	7	0	0	0		
	CDH80	197	0	0	0	0	11	79	90	17	0	0	0	0		
(14) Wind	WSAvg	6.7	6.3	6.4	7.4	7.7	7.7	6.5	5.8	5.8	6.4	7.2	7.0	6.7		
(15) Precipitation	PrecAvg	19.7	0.8	0.7	0.6	1.0	1.7	3.1	2.8	3.1	2.1	1.7	1.3	0.9		
	PrecMax	26.9	1.4	1.7	1.5	2.1	3.2	5.7	6.2	7.0	4.4	2.8	2.8	1.5		
	PrecMin	12.7	0.3	0.1	0.2	0.3	0.6	0.8	0.6	0.8	0.3	0.6	0.6	0.3		
	PrecStd	3.4	0.3	0.4	0.3	0.5	0.7	1.5	1.5	1.6	0.9	0.6	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.8	33.3	44.0	63.2	81.0	86.2	85.6	82.3	73.6	59.2	39.4	32.3		
		MCWB	31.8	31.5	35.9	49.5	60.6	66.9	70.9	68.1	59.5	50.3	37.7	31.2		
	2%	DB	26.3	29.4	39.0	55.0	74.0	82.1	82.6	77.7	67.6	52.6	35.4	29.6		
		MCWB	25.2	28.1	33.1	45.1	56.3	65.8	68.0	65.8	57.4	46.4	34.1	28.6		
	5%	DB	21.7	25.5	36.1	50.2	67.7	78.7	79.7	73.8	63.0	48.7	33.6	26.6		
		MCWB	20.7	23.8	32.2	42.1	54.4	64.7	66.6	63.7	55.2	44.7	32.6	25.6		
	10%	DB	17.2	20.8	33.4	46.0	61.8	74.9	76.6	69.8	58.7	45.2	31.5	22.8		
		MCWB	16.2	19.4	30.5	39.8	51.6	62.9	65.1	61.9	53.1	42.3	30.4	21.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.9	31.8	36.6	50.4	62.2	70.6	73.5	70.1	61.9	51.3	37.8	31.3		
		MCDB	32.9	33.2	42.1	60.4	78.4	80.4	82.8	78.9	69.4	57.9	38.8	31.9		
	2%	WB	25.2	28.3	34.2	46.2	58.3	68.0	70.0	67.2	58.5	47.8	34.4	28.8		
		MCDB	26.2	29.3	37.4	53.0	69.9	78.0	79.1	75.1	65.8	51.2	35.1	29.6		
	5%	WB	20.8	23.9	32.6	43.2	55.5	66.1	68.0	64.9	56.1	45.1	32.8	25.7		
		MCDB	21.7	25.3	35.8	49.6	65.9	76.0	76.7	71.7	61.7	48.2	33.6	26.6		
	10%	WB	16.3	19.4	30.5	39.9	52.3	64.1	66.3	62.6	53.5	42.4	30.4	21.9		
		MCDB	17.2	20.9	33.5	45.6	61.6	72.8	74.2	68.8	58.5	45.0	31.3	22.8		
(35) Mean Daily Temperature Range	MDBR		13.5	14.6	16.7	16.8	16.7	16.4	16.0	15.3	13.3	9.8	11.5	13.0		
	5% DB	MCDBR	14.6	12.5	15.4	20.4	25.4	22.3	20.9	22.0	20.2	14.3	9.0	13.2		
		MCWBR	14.3	11.5	11.6	12.8	12.8	10.5	9.7	11.3	11.4	9.4	8.8	12.9		
	5% WB	MCDBR	14.9	12.6	14.7	19.2	23.4	19.2	18.7	19.4	18.4	13.2	8.9	13.8		
		MCWBR	14.6	11.8	11.9	13.0	12.9	10.3	9.8	10.9	11.2	9.3	8.8	13.6		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.305	0.355	0.377	0.388	0.406	0.447	0.411	0.362	0.330	0.273	0.254		
	taud		1.969	2.002	1.948	2.056	2.275	2.318	2.170	2.266	2.358	2.245	2.023	1.912		
	Ebn at Noon		142	206	233	255	265	261	245	245	238	200	136	91		
	Edh at Noon		15	27	40	43	37	37	42	35	27	21	14	10		
(44) All-Sky Solar Radiation	RadAvg		138	429	912	1410	1539	1741	1616	1145	718	319	136	72		
	RadStd		12	35	57	130	132	144	187	138	91	33	30	7		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page